

Pleistocene Bats from San Josecito Cave, Nuevo Leon, Mexico

J. Knox Jones

Project Gutenberg

Pleistocene Bats from San Josecito Cave, Nuevo Leon, Mexico

J. Knox Jones

Project Gutenberg

The Project Gutenberg eBook of Pleistocene Bats from San Josecito Cave, Nuevo Leon, Mexico

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Pleistocene Bats from San Josecito Cave, Nuevo Leon, Mexico

Author: J. Knox Jones

Release date: October 9, 2009 [eBook #30217]

Most recently updated: October 24, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/30217

Credits: Produced by Chris Curnow, Joseph Cooper, Diane Monico, and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK PLEISTOCENE
BATS FROM SAN JOSECITO CAVE, NUEVO LEON, MEXICO ***

UNIVERSITY OF KANSAS PUBLICATIONS
MUSEUM OF NATURAL HISTORY

Volume 9, No. 14, pp. 389-396

December 19, 1958

**Pleistocene Bats from San Josecito
Cave,
Nuevo León, México**

BY

J. KNOX JONES, JR.

UNIVERSITY OF KANSAS

LAWRENCE

1958

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, Henry S. Fitch, Robert W. Wilson

Volume 9, No. 14, pp. 389-396
Published December 19, 1958

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED IN
THE STATE PRINTING PLANT
TOPEKA, KANSAS
1958

27-5516

Pleistocene Bats from San Josecito Cave, Nuevo León, México

BY

J. KNOX JONES, JR.

Some of the Pleistocene mammals from San Josecito Cave, near Aramberri, Nuevo León, México, collected by field parties of the California Institute of Technology under the direction of the late Professor Chester Stock, have been reported previously (see Furlong, 1943; Cushing, 1945; Stock, 1950; Hooper, 1952; Findley, 1953; Stock, 1953; Handley, 1955; Jackway, 1958). In 1950, Professor Stock loaned a portion of the San Josecito material to the University of Kansas for identification. Included therein were 89 crania and rami of bats, representing three families (Phyllostomidae, Desmodontidae and Vespertilionidae) and five genera, each represented by a single species. One of the species is here described as new. Three of the kinds are known only from the Pleistocene and two are Recent species.

The only previous mention of fossil bats from México known to me concerns material from San Josecito Cave. Cushing (1945:182) mentioned a "vampire bat" from the cave (see also Maldonado-Koerdell, 1948:17), and Handley (1955:48) based his description of *Corynorhinus tetralophodon* on a specimen from San Josecito.

Brief descriptions of the cave have been published by Miller (1943) and Stock (1943). The precise age of the deposits is unknown; stratification data did not accompany the material sent on loan to the University of Kansas. However, most of the micro-fauna is thought to have come from the higher levels in the cave and is probably late Pleistocene.

The San Josecito Cave collections are currently the property of the Los Angeles County Museum.

I am grateful to Dr. E. Raymond Hall for permission to study the bats from San Josecito Cave, to Dr. Robert W. Wilson for criticism of the manuscript,

and to Mr. Philip Hershkovitz for permission to use comparative material at the Chicago Natural History Museum. Lucy Rempel made the drawings from photographs by John M. Legler.

***Leptonycteris nivalis* (Saussure)**

Referred material.—Seventy crania, LACM (CIT) 2951-54, 2956-64, 3114-22, 3124-25, 3127, 3131-35, 3137-41, 3143-55, 3942, 21 unnumbered, of which 35 are nearly complete, lacking zygomatic arches, auditory bullae and some teeth; three rami, one right, LACM (CIT) 3126, and two left, unnumbered.

Remarks.—The long-nosed bats from San Josecito Cave do not differ appreciably from *Leptonycteris nivalis longala* Stains, the largest Recent subspecies of the species, and the subspecies that occurs in the same geographic area today. Average and extremes of three cranial measurements of 22 specimens from San Josecito Cave, followed in parentheses by the average and extreme measurements of 23 adult *L. n. longala* from the type locality, 12 mi. S and 2 mi. E Arteaga, 7500 ft., Coahuila (after Stains, 1957: 356), are: Greatest length of skull, 28.2, 27.2-28.9 (27.5, 26.1-29.0); least interorbital constriction, 5.0, 4.8-5.4 (4.8, 4.1-5.4); breadth of braincase, 11.1, 10.6-11.6 (10.7, 10.1-11.2). The San Josecito specimens average larger than the series of Recent specimens in all of these measurements, especially breadth of braincase, but there is considerable overlap in each case and the extremes of greatest length of skull and of least interorbital constriction do not exceed the extremes in the Recent series.

***Desmodus stocki*, new species**

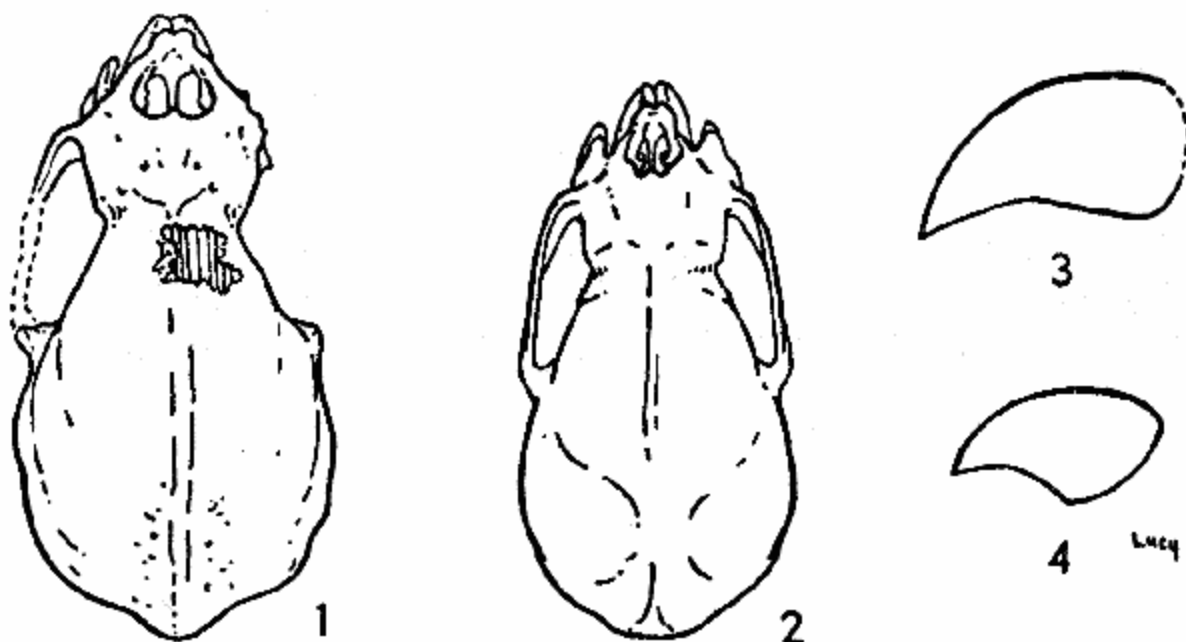
Holotype.—Cranium, lacking post-incisor dentition on the left side, zygomatic arches and auditory bullae; Los Angeles County Museum (CIT) No. 3129; from Pleistocene deposits of San Josecito Cave, near Aramberri, Nuevo León, México.

Referred material.—Twelve additional partial crania, LACM (CIT) 2946-50, 3127-30, 3940-41, 2 unnumbered.

Diagnosis.—Resembling the Recent *Desmodus rotundus* but differing from it as follows: Skull larger (see measurements and [Figs. 1-2](#)), heavier and more massive; rostrum and braincase relatively as well as actually broader, interorbital region relatively more constricted; braincase more rounded (less elongate) as viewed from above; nasals less concave in lateral view; narial vacuity broader in relation to greatest length of skull, more nearly heart-shaped; palate broad, less concave medially; mesopterygoid fossa relatively and actually broader anteriorly, the sides nearly parallel; zygomatic arches (judging from No. 2950, the only specimen with a complete arch, the left) less rounded in outline, appearing broader owing to the more constricted interorbital region.

Dentition larger and heavier than that in *rotundus*, but otherwise differing only slightly from it; upper incisor less concave on cutting surface (see Figs. 3-4); premolar and molar slightly less bladelike, with heavier roots.

The peculiar shape of the incisor of *stocki* is shared to some extent with *Diaemus youngi*, a Recent South American desmodontid. However, *stocki* does not otherwise resemble *D. youngi*, differing from it as follows: Skull larger and heavier; interorbital constriction much narrower; zygomatic arches less strongly bowed; skull less compact, more elongate; braincase and rostrum relatively much narrower in relation to greatest length of skull. Furthermore, specimens of *stocki* show no trace of the minute M2 attributed to *youngi* by de la Torre (Proc. Biol. Soc. Washington, 69: 191, 1956). For cranial measurements of *youngi* see Sanborn (Jour. Mamm., 30: 283, 1949).



Figs. 1-4. Fig. 1. Dorsal view of holotype of *Desmodus stocki*, $\times 1\frac{1}{2}$. Fig. 2. Dorsal view of *Desmodus rotundus murinus*, σ , KU 54969, La Mula, 13 mi. N Jaumave, Tamaulipas, $\times 1\frac{1}{2}$. Fig. 3. Lateral view of left upper incisor of *D. stocki*, LACM (CIT) 2950, $\times 2\frac{1}{2}$. Fig. 4. Lateral view of left upper incisor of *D. r. murinus*, φ , KU 54967, La Mula, 13 mi. N Jaumave, Tamaulipas, $\times 2\frac{1}{2}$.

Remarks.—The essential differences between *D. stocki* and *D. rotundus* are in size and proportion. I do not doubt that the two species are closely related; possibly *stocki* is ancestral to *rotundus*.

The species is named in honor of the late Professor Chester Stock, under whose direction the fossil materials from San Josecito Cave were obtained, and who, at the time of his death, was studying the mammalian fauna from the cave.

***Eptesicus cf. grandis* (Brown)**

Referred material.—One rostrum, with P4-M3 on the right side and P4 only on the left, LACM (CIT) 2990.

Remarks.—This specimen is referred provisionally to *E. grandis*. The dentition is larger and heavier, and the ridges and depressions on the dorsal surface of the rostrum are more pronounced than in Recent *E. fuscus*. The

P4-M3 length is 6.1 (approximately 6.1 in the holotype of *grandis*, less in *fuscus*); least interorbital constriction, 4.2 (4.3 in the holotype of *grandis*, more in *fuscus*); breadth of rostrum between infraorbital canals, 6.4; breadth across P4, 7.3.

TABLE 1.—Cranial measurements of two species of *Desmodus*.

Catalogue number or number of specimens averaged	Greatest length of skull	Condylobasal length	Zygomatic breadth	Breadth of braincase	Least interorbital constriction	Breadth of foramen magnum
<i>Desmodus rotundus murinus</i>, La Mula, 13 mi. N Jaumave, Tamaulipas						
10 (3 ♂, 7 ♀) Ave.	24.3	21.4	12.0	12.1	5.5	5.2
Max.	24.9	22.0	12.5	12.5	5.6	5.3
Min.	23.9	21.0	11.7	11.9	5.3	5.1
<i>Desmodus stocki</i>, San Josecito Cave, Nuevo León						
2946	27.3	24.5		14.2	6.1	5.8
2947				13.6		5.7
2948		24.3		13.9	6.2	5.3
2949		24.7		13.9	6.1	5.5
2950			14.1	13.5		5.7
3127				13.5	6.0	5.7
3128	26.5			13.5	6.2	5.5
3129 (type)	28.2	24.5		13.7	5.9	5.7
3940	27.4	24.4		13.8	6.2	
3941		24.6	14.0	13.7	6.0	5.6

Brown (1908:174) originally named *grandis* as a subspecies of *fuscus*. Gidley and Gazin (1938:11) considered it a distinct species. Whether *grandis* is only a subspecies of *E. fuscus* or a separate species, *grandis* is closely related to *fuscus*, and probably is ancestral to it.

***Lasiurus cinereus* (Palisot de Beauvois)**

Referred material.—One cranium, lacking basioccipital, tympanic and mastoid regions, and most of the dentition, having only M3 on the right side and M2-M3 on the left, LACM (CIT) 3160.

Remarks.—The cranium of No. 3160 is inseparable from those of 10 spring-taken specimens of *L. c. cinereus* from the San Gabriel Mts., Los Angeles Co., California (KU 49727, 49729-37). Measurements of No. 3160, followed by the average and extremes (in parentheses) of the Californian series, are: Condylobasal length, 16.1, 16.5 (15.9-17.2); zygomatic breadth, 12.3, 12.4 (12.0-12.7); least interorbital constriction, 5.2, 5.4 (5.2-5.6); breadth of braincase, 8.7, 9.0 (8.5-9.3); length of palate not including terminal spine, 5.1, 5.3 (4.8-5.9). The teeth of the San Josecito specimen are comparatively unworn. A label with the skull bears the notation "talus" in parentheses, which, in so far as I am able to determine, indicates surface talus inside the cave. Therefore, the specimen in question may be of Recent origin.

It is perhaps worthy of note that *Lasiurus cinereus* is primarily a tree-dwelling bat, although a few Recent specimens have been reported from caves (see Beer, 1954:116).

***Corynorhinus tetralophodon* Handley**

A single cranium of a *Corynorhinus* LACM (CIT) 2989 was included in the original materials sent to Kansas by Professor Stock. Subsequently, this specimen was loaned to Charles O. Handley, Jr., who described it as a new species, *C. tetralophodon*. The latter is said to differ from all other plecotine bats by the retention of a well-developed fourth commissure (ridge extending posteroexternally from metacone) on the M3 (Handley, 1955:48).

LITERATURE CITED

BEER, J. R.

1954. A record of the hoary bat from a cave. Jour. Mamm., 35:116, February 10.

BROWN, B.

1908. The Conard Fissure, a Pleistocene bone deposit in northern Arkansas: with description of two new genera and twenty new species and subspecies of mammals. Mem. Amer. Mus. Nat., 9:155-208, pls. 14-25.

CUSHING, J. E., JR.

1945. Quaternary rodents and lagomorphs of San Josecito Cave, Nuevo Leon, Mexico. Jour. Mamm., 26:182-185, July 19.

FINDLEY, J. S.

1953. Pleistocene Soricidae from San Josecito Cave, Nuevo Leon, Mexico. Univ. Kansas Publ., Mus. Nat. Hist., 5:633-639, December 1.

FURLONG, E. L.

1943. The Pleistocene antelope, *Stockoceros conklingi*, from San Josecito Cave, Mexico. Carnegie Inst. Washington Publ., 551:1-8, 5 pls., February 3.

GIDLEY, J. W., and GAZIN, C. L.

1938. The Pleistocene vertebrate fauna from Cumberland Cave, Maryland. Bull. U. S. Nat. Mus., 171:vi + 99, 50 figs., 10 pls.

HANDLEY, C. O., JR.

1955. *A new Pleistocene bat (Corynorhinus) from Mexico*. Jour. Washington Acad. Sci., 45:48-49, March 14.

HOOPER, E. T.

1952. A systematic review of the harvest mice (genus *Reithrodontomys*) of Latin America. Misc. Publ. Mus. Zool., Univ. Michigan, 77:1-255, 9 pls., 24 figs., 12 maps, January 16.

MALDONADO-KOERDELL, M.

1948. Los vertebrados fosiles del Cuaternario en México. Revista Soc. Mexicana Hist. Nat., 9:1-35, June.

JACKWAY, G. E.

1958. Pleistocene Lagomorpha and Rodentia from the San Josecito Cave, Nuevo León, México. Trans. Kansas Acad. Sci., 61: in press.

MILLER, L.

1943. The Pleistocene birds of San Josecito Cavern, Mexico. Univ. California Publ. Zool., 47:143-168, April 20.

STAINS, H. J.

1957. A new bat (genus *Leptonycteris*) from Coahuila. Univ. Kansas Publ., Mus. Nat. Hist., 9:353-356, January 21.

STOCK, C.

1943. The cave of San Josecito, Mexico. New discoveries of vertebrate life of the ice age. Eng. Sci. Monthly, California Inst.

Tech., Balch Grad. School Geol. Sci. Contrib., 361:1-5,
September.

1950. Bears from the Pleistocene cave of San Josecito, Nuevo Leon,
Mexico. Jour. Washington Acad. Sci., 40:317-321, 1 fig.,
October 23.

1953. El caballo pleistocenico (*Equus conversidens leoni*, subsp.
nov.) de la cueva de San Josecito, Aramberra, Nuevo Leon.
Mem. Congr. Cient. Mex., 3:170-171.

Transmitted August 18, 1958.

27-5516

*** END OF THE PROJECT GUTENBERG EBOOK PLEISTOCENE
BATS FROM SAN JOSECITO CAVE, NUEVO LEON, MEXICO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.